Nptel Online Certification Course: Computer Vision
Week 7: Feature matching and model fitting
Type of questions: MCQ/MSQ
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type: MCQ**

Using the method of least squares, find an equation of the form $y = ax + b$ that fits the data given as (x, y) coordinates. $[(0, 2), (1, 5), (2, 8)]$

a) $y = 3x - 2$

b) $y = 2x + 3$

c) $y = 3x + 2$

d) $y = 3.1x + 1.9$

Correct Answer: c
Detailed Solution: For the least squares fit, $y = ax + b$
The normal equations are:

$$a \sum x_i^2 + b \sum x_i = \sum x_i y_i$$
$$a \sum x_i + nb = \sum y_i$$

Compute the required sums:

x_i	y_i	x_i^2	$x_i y_i$
0	2	0	0
1	5	1	5
2	8	4	16

$$\sum x_i = 3$$
$$\sum y_i = 15$$
$$\sum x_i^2 = 5$$
$$\sum x_i y_i = 21$$
$$n = 3$$

Substituting, $5a + 3b = 21$ and $3a + 3b = 15$
Solving, $a = 3$ and $b = 2$.

QUESTION 2: **Type: MCQ**

Consider two points $a = (2, 5)$ and $b = (4, 1)$ in image space, compute two straight lines corresponding to them in parameter space using Hough transform.

a) $c = -2m + 5$ and $c = -4m + 1$

b) $c = -2m + 1$ and $c = -4m + 5$

c) $c = -5m + 2$ and $c = -m + 4$

d) $c = 2m - 5$ and $c = 4m - 1$

Correct Answer: a
Detailed Solution: From the line equation $y = mx + c$, $c = -mx + y$ is used with given image points to transform them from image space to parameter space.

QUESTION 3: **Type: MSQ**

Which of the following are examples of model fitting given some data points?

a) Estimating an affine transformation matrix between two keypoint sets

b) Image histogram equalized profile optimization matching maps

c) Finding an optimal 3D plane mapping model out of scattered point cloud data arrays

d) Robustly localized circle estimation from noisy boundary point edges

Correct Answer: a, c and d
Detailed Solution: Model fitting identifies optimal parametric descriptions from noisy observed geometric point distributions.

QUESTION 4: **Type: MCQ**

Given two features each of size 4 as $f = [1, 3, 2, 6]$ and $g = [2, 2, 4, 4]$. Compute normalized cross correlation and cosine similarity, respectively.

a) NCC = 0.8944 and Cosine = 0.5345

b) NCC = 0.5345 and Cosine = 0.8944

c) NCC = 0.6000 and Cosine = 0.7500

d) NCC = 0.4444 and Cosine = 0.7071

Correct Answer: b
Detailed Solution: Normalized cross correlation is $\frac{\text{cov}(f,g)}{\sigma_f \sigma_g}$, and cosine similarity is $\frac{f \cdot g}{\|f\|\|g\|}$.

QUESTION 5: **Type: MCQ**

Let d_1 (3), d_2 (9) be the distances to the nearest and the next nearest neighbors from a query feature, respectively for a threshold limit of 0.4 for the method of finding nearest neighbor using nearest neighbor distance ratio (NNDR). Choose the correct option:

a) Nearest neighbor distance ratio d_1/d_2 accepts nearest neighbor as a match

b) Nearest neighbor distance ratio d_1/d_2 rejects nearest neighbor as a match

c) Nearest neighbor distance ratio d_2/d_1 accepts nearest neighbor as a match

d) Nearest neighbor distance ratio d_1/d_2 accepts nearest neighbor only if the ratio is greater than 0.4

Correct Answer: a
Detailed Solution: Nearest neighbor distance ratio d_1/d_2 does not exceed the distance threshold 0.4.

QUESTION 6: **Type: MCQ**

Two probability distributions for 4 discrete values are given as $P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$ and $Q = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$. Compute the KL divergence $D_{KL}(P\|Q)$.

a) 0.3466

b) 0.2500

c) 0.1733

d) 0.3122

Correct Answer: c
Detailed Solution:
The KL divergence is defined as:

$$D_{KL}(P\|Q) = \sum_i p(i) \ln\left(\frac{p(i)}{q(i)}\right)$$

QUESTION 7: **Type: MCQ**

Which of the following statements about a KD-tree is correct?

a) A KD-tree always splits the data using the same coordinate axis at every level.

b) In a KD-tree, each internal node partitions the space into two regions using one coordinate axis.

c) A KD-tree guarantees perfectly balanced partitions irrespective of insertion order.

d) KD-trees can only store 2-dimensional points.

Correct Answer: b)

Detailed Solution:

A K-dimensional tree is a binary search tree used for organizing points in a multi-dimensional space. Each node divides the space into two half-spaces:

Left subtree: points smaller along the active axis

Right subtree: points greater along the active axis

Therefore, option **b)** is correct.

a) Incorrect: KD-trees alternate splitting axes rather than always using the same axis.

c) Incorrect: The balance of a KD-tree depends on insertion order and construction method.

d) Incorrect: KD-trees work for any dimension k, not only 2D.

QUESTION 8: **Type: MCQ**

Using Random Sample Consensus (RANSAC), compute the number of samples N required to achieve a 99% probability (p) of obtaining at least one pure inlier set, given an outlier ratio of 20% (e) and a minimal sample size of $s = 2$. Round to the nearest integer.

a) $N = 5$

b) $N = 7$

c) $N = 9$

d) None of the above

Correct Answer: a

Detailed Solution:

$$N = \frac{\ln(1 - p)}{\ln(1 - (1 - e)^s)}$$

$p = 0.99$, $e = 0.20$, $s = 2$

QUESTION 9: **Type: MCQ**

Consider points in 2D plane (5,5), (12, 10), (8, 1), (2, 8), (10, 14). After building a 2D binary search tree following the given order starting with an X-split, which point becomes the left child of the root node?

a) (12,10)

b) (2,8)

c) (8,1)

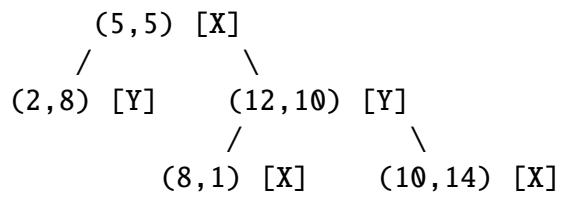
d) (10,14)

Correct Answer: b

Detailed Solution: A 2D KD-tree is constructed by alternating split axes:

$$X \to Y \to X \to Y \to \cdots$$

Root node uses an X-split. Step 1: Insert (5,5). This becomes the root.
Step 2: Insert (12,10). Compare x-coordinate with root. Hence, it goes to the right of $(5, 5)$.
Step 3: Insert (8,1). Compare with root using x-split. Move right to $(12, 10)$. Now compare using y-split. Hence, $(8, 1)$ becomes the left child of $(12, 10)$.
Step 4: Insert (2,8). Compare with root using x-split. Hence, $(2, 8)$ becomes the left child of the root.
Step 5: Insert (10,14). Compare with root using x-split. Move right to $(12, 10)$. Now compare using y-split. $(10, 14)$ becomes the right child of $(12, 10)$.
Final KD-tree:



Therefore, the left child of the root node is $(2, 8)$.

QUESTION 10: **Type: MCQ**

A computer vision researcher uses the method of least squares to fit a straight line $y = ax + b$ to a set of n keypoint tracking coordinates (x_i, y_i). The resulting optimal parameters are found to be $a = 2.5$ and $b = -1.0$.

If the entire image coordinate system is shifted such that every y-coordinate is increased by exactly 3 units (i.e., $y'_i = y_i + 3$), what will be the new slope (a') and intercept (b') for the optimal least squares line fitting the updated coordinates?

a) $a' = 5.5$ and $b' = -1.0$

b) $a' = 2.5$ and $b' = 2.0$

c) $a' = 7.5$ and $b' = -3.0$

d) $a' = 2.5$ and $b' = -1.0$

Correct Answer: b

Detailed Solution: Normal equations of least square fit of straight line is $a\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$ and a $\sum_{i=1}^{n} x_i + nb = \sum_{i=1}^{n} y_i$. Substituting $y'_i = y_i + 3$ changes the right side of the second normal equation to $\sum(y_i + 3) = \sum y_i + 3n$. This systematically increases the intercept parameter by 3 units while leaving the geometric slope unaltered. Thus, $a' = a = 2.5$ and $b' = b + 3 = -1.0 + 3 = 2.0$.
